

Exp No.: 1 Setting up Hyperledger Fabric Development Environment

Aim

To install and configure the Hyperledger Fabric development environment on a Linux system using Docker, Go, and Fabric binaries, create a blockchain test network, and establish a communication channel between peer organizations.

Software Requirements

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Go (Version 1.20 or later)
- Git
- Curl
- Hyperledger Fabric Samples
- Hyperledger Fabric Docker Images

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB free disk space
- Internet Connection

Procedure

Step 1: Update the System

```
sudo apt update
```

```
sudo apt upgrade -y
```

Step 2: Install Required Packages

```
sudo apt install -y git curl jq docker-compose
```

Verify the installation.

```
git --version
```

```
curl --version
```

```
docker-compose --version
```

Step 3: Verify Docker Installation

```
docker --version
```

Start and enable Docker.

```
sudo systemctl start docker
```

```
sudo systemctl enable docker
```

Add the current user to the Docker group.

```
sudo usermod -aG docker $USER
```

```
newgrp docker
```

Verify Docker.

```
docker run hello-world
```

Step 4: Verify Go Installation

```
go version
```

Example Output

```
go version go1.22.0 linux/amd64
```

Step 5: Create a Workspace

Create a directory for the Hyperledger Fabric project.

```
mkdir -p ~/fabric-dev
```

```
cd ~/fabric-dev
```

Verify the directory.

```
pwd
```

Step 6: Download Hyperledger Fabric Samples

Download the Fabric binaries, Docker images, and sample applications.

```
curl -sSL https://bit.ly/2ysbOFE | bash -s
```

Wait until all files are downloaded successfully.

Step 7: Navigate to the Fabric Samples Directory

```
cd fabric-samples
```

Step 8: Configure Environment Variables

```
echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin'
```

```
>> ~/.zshrc
```

```
source ~/.zshrc
```

Step 9: Verify Hyperledger Fabric Installation

Verify the peer binary.

```
peer version
```

Verify the orderer binary.

```
orderer version
```

Expected Output

```
peer:
```

```
Version: v2.5.16
```

```
orderer:
```

```
Version: v2.5.16
```

Step 10: Start the Hyperledger Fabric Test Network

Navigate to the test network.

```
cd ~/fabric-dev/fabric-samples/test-network
```

Start the network.

```
./network.sh up
```

The command performs the following tasks:

- Generates cryptographic certificates •
- Creates Docker containers
- Starts the orderer
- Starts peer organizations
- Creates the Fabric network

Step 11: Verify Running Containers

Display all active Docker containers.

`docker ps`

Expected containers include:

- `orderer.example.com`
- `peer0.org1.example.com`
- `peer0.org2.example.com`

Step 12: Create a Channel

Create an application channel named **mychannel**.

`./network.sh createChannel -c mychannel` This command:

- Generates the genesis block
- Creates the channel
- Joins both organizations
- Configures the anchor peers

Step 13: Verify Successful Channel Creation

Successful messages include:

Channel 'mychannel' created successfully

Successfully submitted proposal to join

channel Anchor peer set for Org1

Anchor peer set for Org2

Step 14: Stop the Network

After completing the experiment, stop the Fabric network.

`./network.sh down`

Flow of the Setup

Install Dependencies



Install Docker & Go



Download Hyperledger Fabric



Configure Environment Variables



Verify Installation



Start Test Network



Create Channel

↓
Join Peer Organizations

↓
Configure Anchor Peers

↓
Hyperledger Fabric Environment Ready

OBSERVATION :

1. Opening Ubuntu

```
dharsshna@DHARSHU: ~/fabr
dharsshna@DHARSHU:~$ peer version
peer:
  Version: v2.5.16
  Commit SHA: fa71cf9
  Go version: go1.26.4
  OS/Arch: linux/amd64
  Chaincode:
    Base Docker Label: org.hyperledger.fabric
    Docker Namespace: hyperledger

dharsshna@DHARSHU:~$ cd ~/fabric-dev/fabric-samples/test-network
dharsshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Using docker and docker-compose
Stopping network
ERROR: Top level object in './addOrg3/compose/docker/docker-compose-couch-org3.yaml' needs to be an object not '<class 'NoneType'>'.
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
Removing generated docker images
dharsshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ ./network.sh up
Using docker and docker-compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb' with crypto from 'cryptogen'
LOCAL VERSION=v2.5.16
DOCKER IMAGE VERSION=v2.5.16
/home/dharsshna/fabric-dev/fabric-samples/test-network/../../bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.yaml --output=organizations
org2.example.com
+ res=0
```

2. Testing Network

```
dharsshna@DHARSHU: ~/fabr
Using docker and docker-compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb' with crypto from 'cryptogen'
LOCAL VERSION=v2.5.16
DOCKER IMAGE VERSION=v2.5.16
/home/dharsshna/fabric-dev/fabric-samples/test-network/../../bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.yaml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer.yaml --output=organizations
orderer.example.com
+ res=0
Generating CCP files for Org1 and Org2
Creating orderer.example.com ... done
Creating peer0.org1.example.com ... done
Creating peer0.org2.example.com ... done
CONTAINER ID        IMAGE               COMMAND                  CREATED              STATUS              PORTS
6d0fbdee32a        hyperledger/fabric-orderer:latest "orderer"              Less than a second ago Up Less than a second 0.0.0.0:7050->7050/tcp, [::]:9443->9443/tcp
368d6d196af9        hyperledger/fabric-peer:latest "peer node start"       Less than a second ago Up Less than a second 0.0.0.0:9051->9051/tcp, [::]:9445->9445/tcp
237d4569563d        hyperledger/fabric-peer:latest "peer node start"       Less than a second ago Up Less than a second 0.0.0.0:7051->7051/tcp, [::]:9444->9444/tcp
dharsshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

3. Starting the Network

```

dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ ./network.sh up
Using docker and docker-compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb' with crypto from 'cryptogen'
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
/home/dharshna/fabric-dev/fabric-samples/test-network/./bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.yaml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer.yaml --output=organizations
orderer.example.com
Generating CCP files for Org1 and Org2
Creating orderer.example.com ... done
Creating peer0.org1.example.com ... done
Creating peer0.org2.example.com ... done
CONTAINER ID        IMAGE                                     COMMAND                  CREATED              STATUS              PORTS
6d0fbdddee32a      hyperledger/fabric-orderer:latest      "orderer"               Less than a second  Up Less than a second    0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp

```

4. Verifying Containers

```

dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ docker ps
CONTAINER ID        IMAGE                                     COMMAND                  CREATED              STATUS              PORTS
6d0fbdddee32a      hyperledger/fabric-orderer:latest      "orderer"               48 seconds ago      Up 47 seconds      0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
369fd6196af9      hyperledger/fabric-peer:latest        "peer node start"       48 seconds ago      Up 47 seconds      0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
2374d569563d      hyperledger/fabric-peer:latest        "peer node start"       48 seconds ago      Up 47 seconds      0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp

```

5. Create mychannel

```

+ res=0
Creating channel mychannel
Adding orderers
+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
Status: 405
{
  "error": "cannot join: channel already exists"
}

Channel 'mychannel' created
Joining org1 peer to the channel...
Using organization 1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=1
2026-07-17 10:18:05.234 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
Error: proposal failed (err: bad proposal response 500: cannot create ledger from genesis block: ledger [mychannel] already exists
th state [ACTIVE])
after 5 attempts, peer0.org1 has failed to join channel 'mychannel'
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$

```

6. Verification

```

dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ ./network.sh createChannel -c mychannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb'
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
peer0.org1.example.com is up-to-date
peer0.org2.example.com is up-to-date
orderer.example.com is up-to-date
CONTAINER ID        IMAGE                                     COMMAND                  CREATED              STATUS              PORTS
6d0fbdddee32a      hyperledger/fabric-orderer:latest      "orderer"               6 minutes ago       Up 6 minutes       0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
50->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
369fd6196af9      hyperledger/fabric-peer:latest        "peer node start"       6 minutes ago       Up 6 minutes       0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
51->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
2374d569563d      hyperledger/fabric-peer:latest        "peer node start"       6 minutes ago       Up 6 minutes       0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
51->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
Using docker and docker-compose
Generating channel genesis block 'mychannel.block'
Using organization 1
/home/dharshna/fabric-dev/fabric-samples/test-network/./bin/configtxgen
+ '[' 0 -eq 1 ']'

```

```

dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ docker ps
CONTAINER ID        IMAGE                                     COMMAND                  CREATED              STATUS              PORTS
6d0fbdddee32a      hyperledger/fabric-orderer:latest      "orderer"               9 minutes ago       Up 9 minutes       0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
50->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
369fd6196af9      hyperledger/fabric-peer:latest        "peer node start"       9 minutes ago       Up 9 minutes       0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
51->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
2374d569563d      hyperledger/fabric-peer:latest        "peer node start"       9 minutes ago       Up 9 minutes       0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
51->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$

```

6. Stopping the network

```
dnarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Using docker and docker-compose
Stopping network
ERROR: Top level object in './addOrg3/compose/docker/docker-compose-couch-org3.yaml' needs to be an object not '<class 'NoneType'>'.
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
6d0fbddee32a
369fd6196af9
2374d569563d
Removing generated chaincode docker images
dnarshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Result

The Hyperledger Fabric development environment was successfully installed and configured. The required Fabric binaries and Docker images were installed, the test network was created successfully, the **mychannel** application channel was generated, both peer organizations joined the channel, and the anchor peers were configured successfully. The development environment is now ready for deploying chaincode (smart contracts) and developing blockchain applications.